Aayaann Kausar

AI Researcher | LLM Post-Training & Agentic AI Engineering
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Summary

IIT Kharagpur AI researcher with 2+ years of research and engineering experience advancing LLM post-training, knowledge retention and self-improving agents. Architected LangGraph systems for autonomous scientific discovery, drawing on PyTorch, PEFT/QLoRA, quantization, efficient inference and multilingual speech AI. Secured 9th Rank worldwide in the Wunder Fund HFT ML challenge.

Education

Indian Institute of Technology (IIT) Kharagpur — Kharagpur, India
Bachelor of Technology | **Minor in Artificial Intelligence** — 2024 – Present

Research Experience

Summer Research Intern | IIIT Bangalore — May 2025 – Jun 2025

Assessed post-training quantization of DistilBERT for network intrusion detection, focusing on CPU inference latency, classification accuracy and FPGA accelerator integration

- Applied **INT8 post-training quantization** to DistilBERT's linear layers; compared FP32 and INT8 results on CICIDS2018
- Measured accuracy of **96% versus 91%** on **500 samples**, with a **2.41x CPU speedup** averaged over 100 timing runs
- Exported **MHA** tensors via PyTorch hooks to **ONNX** and INT8 C headers for integration with a **Vitis HLS FPGA** accelerator

Independent Research | Subspace Feature Preservation | Code | Paper — Feb 2026 – Oct 2026

Tested whether low-rank activation subspaces predict and curb catastrophic forgetting in LLMs while retaining plasticity for unseen tasks

- Identified **32 ablation-tested PCA directions** in **Qwen2.5-1.5B** whose drift predicts forgetting ($R^2 = 0.933$) versus random axes ($R^2 = 0.850$)
- Proposed rank-32 **SFP regularization** (0.754 retention, 0.650 plasticity), attaining a **0.699 harmonic score** versus 0.631 for LwF
- Formalized a **conditional forgetting bound** in **Lean 4** and documented scale-dependent drift predictability in SmolLM2-135M and Qwen2.5-1.5B

Competitions

HFT Market State Forecasting Challenge | Wunder Fund — Sep 2025 – Nov 2025

Forecasted next-step order-book state with a self-pretrained Transformer, tracking out-of-sample R-squared under a single-thread CPU latency budget

- Ranked **9th of 5,000+ competitors** in the global HFT forecasting challenge, achieving final R^2 of 0.3920 (public score 0.3880)
- Pretrained a **4-layer, 8-head Transformer** using **masked autoencoding**, temporal positional encodings and robust median-IQR scaling
- Accelerated **single-core CPU inference** via **graph compilation**, Numba JIT and preallocated buffers for low-latency streaming forecasts

Projects

Nova | Autonomous AI Co-Scientist | Code — July 2026 – Aug 2026

Developed a multi-agent assistant to propose testable biomedical hypotheses, compare candidates and revise proposals using scientific literature

- Architected **9 LangGraph agents** with **async job queues** for literature search, candidate generation, critique and evolution
- Linked **corrective RAG** and ChromaDB with **Elo tournaments**, semantic diversity checks and feedback-led **proposal revision**
- Generated **10 liver-fibrosis hypotheses** in **11.5 minutes** across **105 model calls**, with LangSmith tracing and human approval gates

NexusAgent | Enterprise Decision Intelligence Platform | Code — Aug 2026 – Sep 2026

Connected hybrid document retrieval with read-only SQL queries, using agent routing to return answers with supporting evidence

- Optimized hybrid retrieval with **Milvus**, BM25, RRF and BGE reranking, lifting Hit@1 from **85.62% to 96.25%** and MRR@5 to **98.12%**
- Hardened MCP NL2SQL with **SQLGlot** AST validation, read-only scopes and query timeouts, passing all **28 security boundary tests**
- Verified 3 LangGraph agent routes and grounded evidence-review logic across **1,941 unit tests** and **323 offline integration tests**

Real-Time Voice Translator | Multilingual Speech AI | Demo | Code — Sep 2026 – Oct 2026

Engineered a streaming speech-to-speech translator to support live bilingual conversations across seven spoken languages

- Built a streaming ASR-MT-TTS pipeline for bidirectional **7-language** translation with Whisper, GPT-OSS 20B and ElevenLabs Flash
- Implemented **20 ms RMS voice activity detection** with **600 ms endpointing**, polyphase resampling and stale-response filtering
- Deployed a **Next.js AudioWorklet** client and FastAPI backend on Vercel and Render, with **p50/p95 latency telemetry** and **27 automated tests**

Coursework

Mathematics: Advanced Calculus | Probability and Statistics | Linear Algebra | Numerical Analysis

CS: Data Structures and Algorithms **AI:** Deep Learning Foundations | Reinforcement Learning | Foundations of LLMs

MOOCs: Language Modeling from Scratch (**Stanford CS336**), Self-Improving AI Agents (**Stanford CS329A**)

Skills & Expertise

ML & LLM Training: Python, PyTorch, TensorFlow, scikit-learn, XGBoost, HF Transformers, PEFT/QLoRA, TRL, CUDA, Triton
Agent Orchestration & RAG: LangGraph, MCP, Multi-Agent Workflows, Milvus, ChromaDB, LangSmith, SQLGlot, asyncio
Deployment & Infrastructure: Linux, Docker, FastAPI, ONNX Runtime, INT8 Quantization, SQL, PostgreSQL, Git, C++, Bash

Scholastic Achievements

JEE Main 2024 | Top **0.29%** of 1.4M+ candidates in the national entrance examination for engineering college admissions
JEE Advanced 2024 | Top **3.2%** of 250K+ aspirants in the national engineering entrance examination for admission to the IITs
WBJEE 2024 | Top **0.20%** among 113K+ candidates in the state entrance examination for undergraduate engineering admissions